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Risk management optimization framework for the optimal deployment of carbon capture and storage system under uncertainty

Shuai Zhang[a], Yu Zhuang[a,b], Linlin Liu[a,*], Lei Zhang[a], Jian Du[a]

[a] Institute of Chemical Process Systems Engineering, Dalian University of Technology, Dalian, 116024, Liaoning, China
[b] Key Laboratory of Liaoning Province for Desalination, School of Energy and Power Engineering, Dalian University of Technology, Dalian, 116024, Liaoning, China

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ABSTRACT

In order to meet the CO_2 reduction target and realize cleaner production, large emission sources have two options: investing in carbon capture and storage (CCS) system and/or paying surcharge for carbon tax. CCS system requires industry to make huge investment in infrastructure and the cost of making improper decisions on CCS is substantial. Many factors can impact the successful deployment of commercial-scale CCS, including significant uncertainty regarding carbon policy, technology and engineering performance. In this study, a two-stage stochastic mixed-integer linear programming (MILP) model is formulated to explore the relationship between these two reduction options by minimizing expected total cost (ETC) for CO_2 reduction, considering carbon tax uncertainty. Furthermore, to assess the risk imposed by the uncertainty, three financial risk metrics are introduced for risk management and each metric is formulated as secondary objective to ETC in a multi-objective optimization model with two objectives. The trade-offs between the economic and risk objectives are obtained via the ε-constraint method. At last, a case of power plants in Northeast China is studied. Results show that the risk-neutral solution may confront high risk or even be infeasible in some extreme scenarios, and the carbon tax price of no less than $ 50/t is recommended for policy-makers. In risk-management cases, to control the risk, variability index metric is reduced from $ 40 $\times 10^8$ to $ 0, probability financial metric is reduced from 0.20 to 0.12 and downside risk metric is reduced from $ 27.09 $\times 10^8$ to $ 13.38 $\times 10^8$ at the expense of increasing ETC.

1. Introduction

Due to the high level of global energy consumption, demand for energy supply is predicted to increase significantly over the next decades [1]. An obvious consequence of this situation is the increased use of fuels, especially traditional carbon-rich sources (i.e., coal, oil and natural gas) that have provided majority of the energy since the industrial revolution [2]. However, with the increasing usage of fossil fuels, the environmental problem of "greenhouse effect" is worsening [3]. Fossil fuels combustion accounts for the largest share of carbon dioxide (CO_2), a primary anthropogenic greenhouse gas (GHG) [4]. Recently, CO_2 emission in the atmosphere is expected to increase at an average rate of 0.6% from 2015 to 2040 [5]. Therefore, reducing CO_2 emission resulting from fossil fuel combustion is of great importance for environmental sustainability. Aimed at mitigating the global mean temperature rise, the 21st United Nations Climate Change Conference (COP21) makes a commitment that the global temperature rise in this century should be limited to no more than 2 °C [6].

Carbon capture and storage (CCS) is a technology that has received industries as well as government attention in CO_2 reduction as it has a number of advantages for cleaner production [7]. First, CCS is a technically feasible method for the decarbonization of energy intensive industries such as power plants, iron and steel manufactures, cement production, and oil refining [8]. According to the IEA's BLUE Map scenario, CCS could achieve for 19% of the electricity related reduction by 2050 [9]. Second, given that the fossil fuels are still expected to constitute backbone of energy mix in the foreseeable future [10], CCS enables the continued use of fossil fuels without imposing rapid transition to the renewable energy system. Finally, CCS can be integrated into low-carbon or carbon-neutral bioenergy carbon capture and storage (BECCS) system to generate negative emission [11]. As a safe and efficient method of preventing CO_2 from entering into atmosphere, CCS plays a crucial role in smoothing the transition to truly sustainable energy production from renewable sources.

* Corresponding author. Room D-408, Chemical Engineering Department, Dalian University of Technology, No.2 Linggong Road, Ganjingzi District, Dalian, Liaoning Province, 116024, China.
E-mail address: liulinlin@dlut.edu.cn (L. Liu).

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Nomenclature

Abbreviations

BECCS	bioenergy carbon capture and storage
CCS	carbon capture and storage
CCUS	carbon capture utilization and storage
CO_2	carbon dioxide
COP21	21st conference of parties
ETC	expected total cost
GHG	greenhouse gas
IEA	international energy agency
MILP	mixed-integer linear programming
MINLP	mixed-integer nonlinear programming
O&M	operating and maintenance
PF	perfect foresight
ROA	real option approach

Sets

I	set of CO_2 emission source
J	set of CO_2 storage sites
R	set of standard pipelines with specified diameter
S	set of all the generated scenarios by sampling method
T	set of time intervals throughout the planning horizon

Parameters

C_p	material price, $/kg
E_f	longitudinal joint factor for the pipeline
E_i	total CO_2 emission available at site i, Mt/y
FC_i^c	capital cost of a carbon capture facility for power plant i, $
FC_r^p	capital cost of a pipeline with diameter r, $
FC_j^s	capital cost of a carbon storage facility for storage site j, $
F_f	design factor for the pipeline
f_m	fraction factor denotes the fraction of material cost
I	capital cost for pipeline
L	the length of pipeline, km
M	auxiliary parameter for the Big-M constraint
OC_i^c	operating cost of a carbon capture facility for power plant i, $/t CO_2
OC_r^p	operating cost of a pipeline with diameter r, $/t CO_2
OC_j^s	operating cost of a carbon storage facility for storage site j, $/t CO_2
P_{max}	maximum operation pressure for the pipeline, MPa
p_s	occurrence probability of scenario s
P_{tax}	price (tax) per ton of CO_2 emission to the atmosphere, $/t CO_2
$Q_r^{p,max}$	CO_2 flow rate upper bound for pipeline with diameter r, Mt/y
$Q_r^{p,min}$	CO_2 flow rate lower bound for pipeline with diameter r, Mt/y
S_j	injection limit for storage site j, Mt/y
S_p	minimum yield stress for the pipeline, MPa
t_p	wall thickness of pipeline, mm
Y_{it}	binary parameter to denote the operation state for power plant i in period t
Y_{jt}	binary parameter to denote the operation state for storage site j in period t
$\rho_{material}$	the density of pipeline material, kg/Nm3
Ω	the maximum cost threshold defined by the decision-maker, $

Variables

DownRisk	downside risk objective for risk management, $
ETC	the expectation for all the costs of scenarios, $
F_{ijts}	amount of CO_2 sent from source i to j in period t for scenario s, Mt
FixedCost	fixed capital cost for all the facilities in the CCS system, $
OperationCost	operating cost for all the facilities in the CCS system, $
ProRisk	probabilistic financial risk objective for risk management
TC_s	total cost of the CCS system for scenario s through the planning horizon, $
VarIndex	variability index objective for risk management, $
xit	1 if a carbon capture facility is installed at source i in year t, 0 otherwise
yit	1 if a carbon storage facility is installed at storage site j in year t, 0 otherwise
zijrt	1 if a pipeline with diameter *r* is installed from source *i* to sink *j* in period *t*, 0 otherwise

In general, CCS comprises three main steps: CO_2 capture from large point sources, CO_2 transportation and CO_2 storage. CO_2 capture technologies can be divided into three categories as depicted in Fig. 1, namely, post-combustion capture, pre-combustion capture and oxyfuel combustion capture [12]. Compared with the other two capture technologies, post-combustion capture is widely applied by the current CCS projects in operation since it is easier to implement. The captured CO_2 is typically compressed at supercritical state for transportation. CO_2 can be transported to the storage sites for permanent sequestration via pipeline, truck, rail and ship. Pipeline has advantage of continuous operating manner in large volume transportation over the other options [13]. The potential storage sites include saline aquifers, depleted oil and gas reservoirs and deep oceans. Several CCS commercial-scale projects are currently under operation throughout the world, for example, Gorgon facility in Australia [14], Jingbian CCS project in China [15] and European CCS infrastructure [16].

The high cost of CCS infrastructure has somewhat hindered its large-scale deployment, so carbon tax approach is proposed for CO_2 emission control from economic incentives. It refers that the government imposes a surcharge on GHG emission for industrial processes [17]. Pricing carbon emission has been used in many countries to discourage CO_2 emission, such as Sweden, Norway and Denmark [18]. In this condition, CO_2 emitters have to decide whether it is beneficial for CO_2 emission sources to install CCS system and/or to pay the surcharge for CO_2 emission. It is hard for decision-makers to perfectly predict the environmental policies, meanwhile considering the fluctuations of carbon tax price. Therefore, the interaction between CCS installation and economic incentives becomes quite uncertain.

In this light, this article aims to fill in the research gaps and incorporate the effect of such uncertainty factor into the CCS planning problem. The CCS planning problem can be considered as a CO_2 disposal problem, where the large point sources emit CO_2 and the storage sites sequestrate CO_2. Since carbon tax prices have direct impact on the design of CCS network design [19], this paper presents a risk management framework by employing three risk metrics to investigate the holistic understanding of the interaction between carbon tax policy and CCS installation under such uncertainty. The proposed framework decides whether it is beneficial for the emitters to be involved into a CCS system and/or pay surcharge for CO_2. The method could provide some quantitative information about the risk imposed by carbon tax fluctuations and inform better policy decisions with insights from a real case study in Northeast China.

The remainder of this article is organized as follows. In Section 2, a review of the literature on CCS planning is presented. Section 3

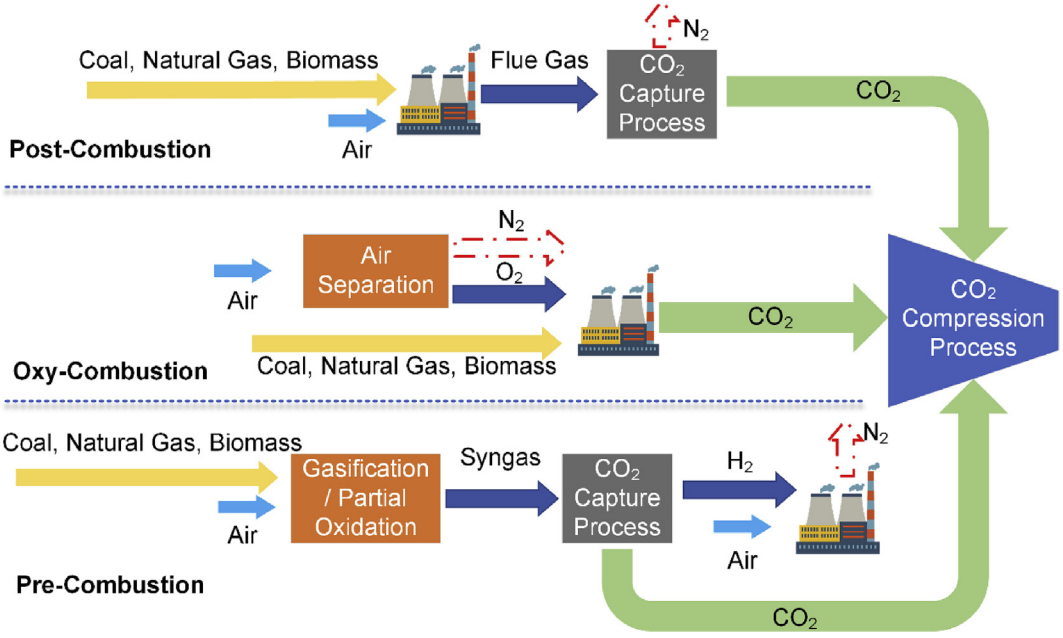


Fig. 1. Three kinds of carbon capture technologies [12].

introduces the proposed risk management methodology. Section 4 shows the detailed mathematical formulations. In Section 5, the case study of power plants in Northeastern China is implemented to illustrate the applications of the proposed models and solution approach. Section 6 discusses the results and comparisons of three risk metrics. Section 7 proffers prospects on practical implications. Finally, Section 8 provides several conclusions and future outlooks for CCS planning problem.

2. Literature review

There has been an extensive amount of research employing mathematical programming to optimize both deterministic and uncertain CCS planning problems. For example, Tan et al. [20] present a continuous-time mixed integer nonlinear programming (MINLP) model with the consideration of injection limit, reservoir capacity and temporal constraints for an optimal CCS network. The objective of their model is to maximize the reduction of CO_2 emission including both captured CO_2 and CO_2 emission of compensatory power generation. To guarantee the global optimality, this model is linearized into a mixed integer linear programming (MILP) model by replacing the bilinear terms. Then, a discrete-time MILP model is presented by Tan et al. [21] for addressing the optimal CO_2 source-sink matching problem. The temporal aspects of source-sink matching are considered because the operating lives of sources and sinks may not completely overlap, especially when the storage sinks are not immediately ready for deployment. Unlike the CO_2 source-sink matching models, a unified MILP model capable of optimizing both carbon capture retrofit and source-sink matching issues simultaneously is developed by Lee et al. [22]. This model unifies the problems of grid power capture retrofitting and the subsequent CO_2 source-sink matching within a single optimization framework. Furthermore, they perform sensitivity analysis to show the interaction between carbon emission and power losses in the system. Although these aforementioned studies are not meant to be a detailed CCS infrastructure planning tool, their hypothetical case studies provide great insights for the early-stage planning of CCS.

Some papers focus on a specific region or country, mostly investigating CCS network in terms of economic optimization. For instance, Middleton and Bielicki [23] propose a comprehensive MILP model named SimCCS that guides decision-maker to economically optimize the spatially dispersed CO_2 sources and sinks. This model can determine where to retrofit CO_2 sources with capture technology, where to store CO_2 through a dedicated pipeline network topology and the quantity of CO_2 to be transported. They show the application of the SimCCS model through a case study in California to highlight the importance of regional CCS planning by examining the sensitivity of CCS infrastructure. Sun and Chen [24] present the improved ChinaCCS Decision Support System (DSS) for optimizing CO_2 pipeline network by generating a cost-minimizing CCS system. This ChinaCCS DSS assists in determining the sources, sinks, amounts for CO_2 captured and storage, as well as the paths and sizes of pipeline. The features of the proposed approach are illustrated through a detailed case study for Beijing–Tianjin–Hebei Region in China. Then, Sun and Chen [25] develop a multi-stage MILP model for source-sink matching within ChinaCCUS DSS. The Beijing–Tianjin–Hebei region in China is still chosen as a case study to address the application of the updated model. They find that the multi-stage model could result in better pipeline connectivity compared to the single-stage model.

More recently, the concept of thorough carbon capture utilization and storage (CCUS) supply chain is proposed by Hasan et al. [26]. They regard the network of CCUS as a supply chain of CO_2 where various industrial CO_2 emission sources supply CO_2, and demands of CO_2 come from various utilization and sequestration sites. Then, Hasan et al. [27] propose a multi-scale framework for the optimal supply chain design for CCUS in the United State, which requires decision making at material, process and supply chain levels simultaneously. They formulate a MINLP model to address the supply chain optimization problem and then simplify the model to a MILP model by assuming that up to 90% of a selected CO_2 source is available. Analogously, Ravi et al. [28] address the supply chain problem by optimizing appropriate sources, capture technologies, pipeline network and CO_2 storage sites for a minimum overall cost to achieve nationwide CO_2 reduction target in the Netherlands. Five different scenarios are formulated by varying the levels and industrial activities of source and storage sites. d'Amore et al. [29] introduce a spatially explicit, multi-echelon and time-dependent MILP model for the economic optimization of a European CCS supply chain for a 20 years' planning horizon. Their results indicate the good European potential for carbon sequestration and give some indications on the total cost for CCS in Europe.

The aforementioned studies rely on deterministic optimization

method without consideration of uncertainty, which assume that all the parameters are known in advance. However, although the technologies of the CCS system have been widely used for demonstration projects, large-scale CCS planning is still lack of enough information, which will introduce uncertainty into CCS planning [30]. The impacts of various uncertainty elements on planning CCS are profound. Multiple factors will affect the success of commercial-scale CCS infrastructure including inherent uncertainties associated with CO_2 emission sources (e.g., source operating lifetime), capture technologies, and CO_2 geological reservoirs (e.g., sink storage capacity) [31]; and external factors such as carbon policy, technology, engineering performance and market forces [30].

The effects of uncertainty in geological reservoir have been widely addressed in terms of site screening [32], permeability and porosity of reservoir [33] and engineering performance [34]. Han and Lee [35] propose a comprehensive CCS infrastructure model with the consideration of uncertainty arising from the fluctuation of CO_2 emission level. The capability of their inexact model is tested by an industrial region on the eastern coast of Korea in 2020. Then they extend this method by developing a refined set of scenarios and considering other uncertainties such as operating cost and product cost [36]. In their study, uncertainties are assumed to follow a set of scenarios associated with given probabilities of occurrence. These uncertainties are modeled as the coefficients of the objective function using a multi-scenario stochastic programming approach. While He et al. [37] address the robust optimal source-sink matching under uncertainty of operation time of sources and sinks. Based on the deterministic MILP model, they develop two uncertain MILP models: worst-case model and robust model. Their results indicate that the robust MILP model provides a more practical solution alternative. Knoope et al. [38] analyze the influence of uncertainty on the layout and cost of a CO_2 network. To achieve this, the cases with and without uncertainty are studied by the perfect foresight (PF) model and real option approach (ROA) model respectively. The uncertainties are included in the CO_2 price, tariff of CO_2, time difference and probability moment that sources join the transportation network. The results show that uncertainty may lead to a less integrated CO_2 infrastructure network with many point-to-point pipelines. Lee et al. [39] propose a stochastic programming model to optimize CCS supply chain in South Korea, aiming to evaluate the interaction among cost, environmental impact and risk. Their results show that uncertainties greatly affect the CCUS network configuration, as well as the optimal CCS operating conditions.

In addition to technological uncertainty, uncertainty in carbon policy has prompted industrial producers to observe government's attitude toward to CO_2 mitigation [40]. Zhang et al. [41] propose a game-theory based framework that integrating CCS infrastructure and cap-and-trade scheme in Qatar. In this framework, the fair design of the CCS infrastructure is obtained under two fairness scenarios: the same saving ratio approach and the game theory Nash approach. They conclude that the cost distributions among power plants vary resulting from the selected CCS system, carbon trading prices and transfer amounts between power plants. Although significant progress has been achieved in optimization of CCS planning, few studies address the relationship between carbon tax pricing and installation of CCS system. What's more, previous studies can only provide risk-neutral solutions by disregarding decision-maker's attitude towards the risk associated with uncertainty. To properly control the risk caused by uncertainty, a risk management tool should be incorporated within the planning of CCS system.

Following these considerations, the goal of this work is to propose an optimization framework evaluating the relationship between the two CO_2 reduction strategies: the carbon tax on CO_2 emission and the planning of CCS system. The uncertainty in carbon tax pricing and its impacts on the deployment of CCS system are modeled and analyzed. To the best of the authors' knowledge, there is no research has studied the risk problem on CCS system, especially the risk imposed by the uncertainty of carbon tax policy. Thus, a two-stage stochastic MILP model is developed first to simultaneously address CCS capital decisions, CO_2 transportation operating decisions, as well as CO_2 source-sink matching design under carbon tax uncertainty. The proposed model decides whether it is beneficial for the CO_2 emission sources to install CCS system and/or to pay the surcharge for the CO_2 emission under carbon tax fluctuation, however, neglects the risk imposed by the uncertainty (e.g., risk-neutral solution). To control the risk, multi-objective models that minimize the expected total cost (ETC) and the risk incorporating with three kinds of risk metric are developed. To solve the multi-objective models and obtain the Pareto-optimal curves respecting to each risk metric, ε-constraint method [42] is employed in this work. A case study in Northeast China is analyzed to demonstrate the effectiveness of the developed models. The obtained results involving risk management are compared with those obtained from risk-

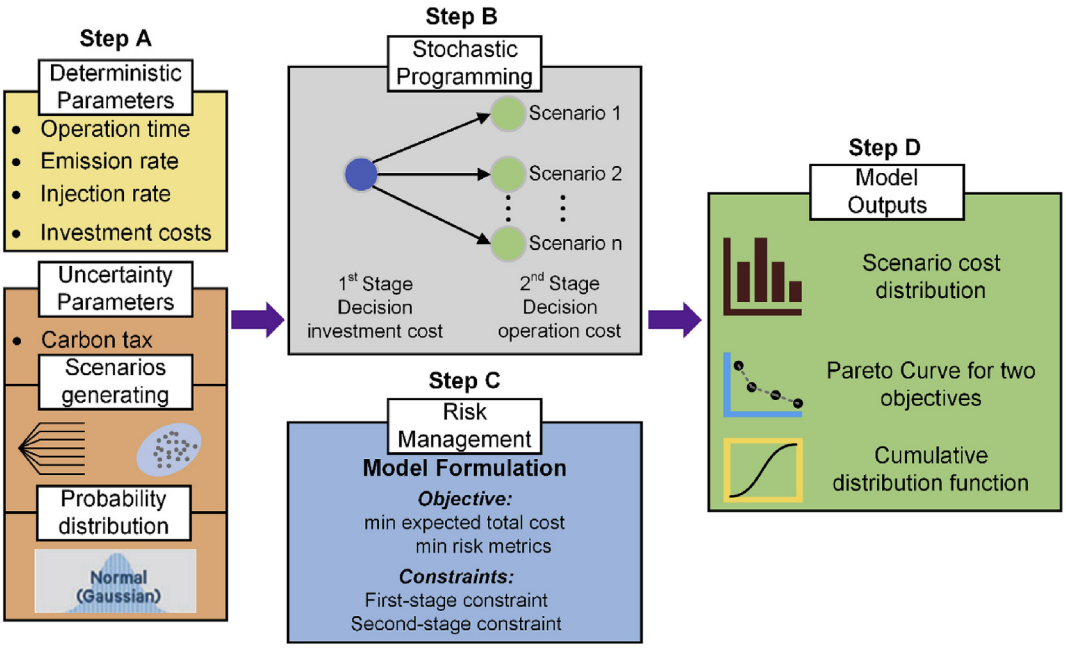


Fig. 2. The proposed optimization framework for CCS planning under uncertainty.

neutral stochastic programming model.

The main contributions of this article are summarized as follows.

i) The interaction between carbon tax pricing and CCS deployment under uncertainty is investigated;
ii) A two-stage stochastic programming model is developed considering the operating lives of emission sources and storage sinks;
iii) Three financial risk metrics-based optimization are applied to account the risk resulted from uncertainty;
iv) A realistic case study of Northeast China region is presented to illustrate the effectiveness of the proposed method and provide quantitative results for the real application of CCS.

3. Methodology

The objective of this work is to develop an optimization-based risk management framework to address the relationship between CCS system and carbon tax under uncertainty, aiding government and industry to make optimal reduction decisions. This is of great significance because the policy uncertainty is one of the multiple factors that impact the success of CCS. The CCS planning scheme without consideration of such uncertainty may confront high risk and cannot be realized in practical operation, which costs a lot for making such wrong decisions on CCS planning. Furthermore, there is still a lacking of quantitative models for CCS risk management, since most of the previous studies are based on qualitative approaches. The proposed risk management framework can provide quantitative information into the decision on how to plan the CCS network while confront low risk level. Another main novelty of this framework is that it can be easily extended to include other uncertain factors if these factors can be depicted by specific probability distribution functions.

The proposed risk management framework (see Fig. 2) can be summarized as follows:

In Step A, it is essential to formulate a deterministic model to address the CCS planning problem, which is taken in Step B as a basis to develop the stochastic formulation. Thus, the deterministic parameters of the problem, such as operation life of CO_2 sources and sinks, CO_2 emission rate and injection rate, and cost-related data (e.g., capital and operating cost for capture, transport, and storage facilities) are needed. The carbon tax pricing is selected as the uncertain parameter to investigate the impacts of this kind of uncertainty on CCS network. To express the realization of uncertainty, this study assumes that uncertain parameters follow normal distribution.

In Step B, a two-stage stochastic MILP model based on deterministic model is developed to simultaneously address CCS investment decisions, CO_2 transportation operating decisions, as well as CO_2 source-sink matching design under uncertainty. The stochastic model is constructed by slightly modifying the deterministic described before with uncertain variables. The objective of this stochastic model is slip into a deterministic term representing decision cost at first-stage and the expected operation cost of a stochastic term which depends on the realization of uncertain parameters at the second-stage. This stochastic model neglects the risk imposed by the uncertainty and thus yields risk-neutral solutions.

In Step C, considering the risk imposed by uncertainty, the associated risk management models based on various risk metrics are presented. These models are formulated as multi-objective models whose objectives are minimum ETC and minimum risk. Herein, three risk metrics (i.e., variability index, probabilistic financial risk, downside risk) with different concerns are applied for risk evaluation in this study. Then, ε-constraint method is employed to solve the risk management models and obtain Pareto-optimal curve respecting to each risk metric. In this way, the multi-objective problem is converted into a single-objective problem.

In Step D, the optimization results (e.g., cost distribution, cumulative distribution curve) can be obtained by assigning the given deterministic and uncertain parameters to the formulated models. The stochastic programming model can provide insights on the relationship between carbon tax and CCS under uncertainty. Solving the risk management models can obtain the Pareto curves of the economic and risk objective and provide quantitative information for decision-makers.

4. Mathematical formulations

In this section, a two-stage stochastic programming model for CCS system design and its extensions for risk management are presented. The models follow the basic construction of two-stage stochastic programming approach, in which decision variables are classified into first-stage (also known as 'here-and-now variables') and second-stage variables (also known as 'wait-and-see variables'). Commonly, the first-stage decision variables refer to design aspects of the problem, which must be made prior to the realization of the uncertain parameters. While the second-stage decision variables refer to operation or recourse decisions of the problem, which are to be determined after the uncertain parameters reveal themselves. In this paper, these decision variables are summarized as follows:

i) First-stage design variables: design aspects of CCS planning, such as where to capture CO2, where to storage CO2, and the pipeline size.
ii) Second-stage operation variables: operating aspects of CCS planning, such as how much to capture and store CO2, capture operation time, and storage schedule.

The two-stage stochastic programming model without considering any risk metrics is presented in section 4.1. Based on two-stage stochastic programming model, the risk management models are presented in section 4.2.

4.1. Two-stage stochastic programming model

In this section, the two-stage stochastic programming model is introduced. It is assumed that the uncertainty associated with carbon tax is characterized by a finite number of scenarios with specified probability. These scenarios are sampled from predefined probability distribution function randomly using Monte Carlo method. A two-stage stochastic programming model is formulated on the base of Zhang et al. [43] to tackle the optimal deployment of CCS system under carbon tax uncertainty. Variables x_{it}, y_{jt} and z_{ijrt} are classified as first-stage variables, which indicate the installation of capture facilities, storage facilities and selection of pipelines. Variable F_{ijts} is assigned to second-stage variable, which represents the amount of the CO_2 to be transported from source i to sink j. Mathematically, the two-stage stochastic MILP model can be formulated as follows:

Objective function:

$$ETC = E[TC_s] = FixedCost + \sum_{s\in S} p_s \times OperationCost_s \tag{1}$$

$$FixedCost = \sum_{i\in I} FC_i^c x_{it} + \sum_{j\in J} FC_j^s y_{jt} + \sum_{i\in I}\sum_{j\in J}\sum_{r\in R} FC_r^p z_{ijrt} \tag{2}$$

$$OperationCost_s = \sum_{t\in T}\left[\sum_{i\in I} OC_i^c F_{ijts} + \sum_{j\in J} OC_j^s F_{ijts} + \sum_{i\in I}\sum_{j\in J}\sum_{r\in R} OC_r^p F_{ijts} + p_{tax}\left(T - \sum_{i\in I}\sum_{j\in J} F_{ijts}\right)\right] \quad \forall s \in S \tag{3}$$

The objective function of this model is to minimize the ETC, which is the expectation of total cost for all scenarios. The objective is comprised of two parts, the fixed capital cost for facilities, *FixedCost*, as shown in equation (2), and the expectation of operating and

maintenance (O&M) cost for all scenarios. The O&M cost for each scenario is shown as equation (3). In equation (2), the capture facility capital cost, storage facility capital cost and pipeline capital cost are included. The binary variables x_{it}, y_{it}, and z_{ijrt} are used to denote the installment of capture facility, storage facility and pipeline in year t and these variables are classified as the first-stage variables, which mean that they must be determined prior to the realization of uncertainty parameters. $OperationCost_s$ in equation (3) is the operating and maintenance cost for carbon capture, storage, pipeline transportation and surcharge cost for scenario s, depending on the amount of carbon captured in period t, F_{ijts}. Variable F_{ijts} is assigned to second-stage variable. Since the uncertainty parameters vary with the discrete scenarios, the expected O&M cost equals to the product of the scenario probability, p_s, and the associated O&M cost, $OperationCost_s$, summed over all the scenarios s. Towards this objective function, the optimal CCS deployment under uncertainty is optimized subjecting to the constraints as follows:

Considering the practical factors in pipeline operation such as pressure drop, erosion speed, etc., the amount of CO_2 sent through each pipeline in period t should be restricted within a certain capacity level once the pipeline with a certain diameter is constructed. So, equations (4) and (5) are given to ensure these limits by using binary z_{ijrt} to indicate the installment of the pipeline, $Q_r^{p,\max}$ and $Q_r^{p,\min}$ to present the upper and lower bounds of the capacity respectively.

$$F_{ijts} \le \sum_{r\in R} Q_r^{p,\max} z_{ijrt} \quad \forall\, \mathrm{i}\in \mathrm{I},\mathrm{j}\in \mathrm{J},\mathrm{t}\in \mathrm{T},\mathrm{s}\in \mathrm{S} \tag{4}$$

$$F_{ijts} \ge \sum_{r\in R} Q_r^{p,\min} z_{ijrt} \quad \forall\, \mathrm{i}\in \mathrm{I},\mathrm{j}\in \mathrm{J},\mathrm{t}\in \mathrm{T},\mathrm{s}\in \mathrm{S} \tag{5}$$

Equation (6) indicates that only one pipeline with a certain diameter can be constructed to connect the CO_2 source i and sink j.

$$\sum_{j\in J}\sum_{r\in R} z_{ijrt} \le 1 \quad \forall\, i\in I,\, t\in T \tag{6}$$

where z_{ijrt} is a binary variable, which represents that if a supercritical CO_2 pipeline with diameter r is constructed at period t between source i and sink j, the value is 1. If there are existent pipelines already, they are useable by setting z_{ijrt} of that certain pipeline to 1.

Equation (7) indicates that the amount of CO_2 captured in emission source i during period t cannot exceed the maximum emission (E_i) of that source i. x_{it} is a binary variable, which is equal to 1 if the carbon capture facility at emission source i operates in period t.

$$\sum_{j\in J} F_{ijts} \le E_i \mathrm{x}_{it} \quad \forall\, \mathrm{i}\in \mathrm{I},\mathrm{t}\in \mathrm{T},\mathrm{s}\in \mathrm{S} \tag{7}$$

Similar to the constraint for emission source, equation (8) ensures that the amount of CO_2 stored at a sink j in period t is no more than the predefined injection rate (S_j) of that sink. y_{jt} is a binary variable, which is equal to 1 if the carbon storage facility at sink j operates in period t.

$$\sum_{i\in I} F_{ijts} \le S_j y_{jt} \quad \forall\, \mathrm{j}\in \mathrm{J},\mathrm{t}\in \mathrm{T},\mathrm{s}\in \mathrm{S} \tag{8}$$

Equation (9) is introduced to ensure the operation time of capture facility x_{it} and the life time of the emission source are overlapped. By this constraint, it is able to stipulate that none capture is available if an emission source i is not in operation in that year. Similar to the emission source, none CO_2 is stored if that storage sink j is not open in period t, as presented in equation (10).

$$\mathrm{x}_{it} \le Y_{i,t} \quad \forall\, \mathrm{i}\in \mathrm{I},\mathrm{t}\in \mathrm{T} \tag{9}$$

$$y_{jt} \le Y_{\mathrm{j,t}} \quad \forall\, j\in J,\mathrm{t}\in \mathrm{T} \tag{10}$$

where $Y_{i,t}$ and $Y_{j,t}$ are binary parameters to denote the start time and end life time of the source i and storage sink j.

Equations (11)–(13) specify that these variables are binary variables and can only be equal to 0 or 1.

$$x_{it} \in \{0,\ 1\} \quad \forall\, \mathrm{i}\in \mathrm{I},\mathrm{t}\in \mathrm{T} \tag{11}$$

$$\mathrm{y}_{jt} \in \{0,\ 1\} \quad \forall\, \mathrm{j}\in \mathrm{J},\mathrm{t}\in \mathrm{T} \tag{12}$$

$$z_{ijrt} \in \{0,\ 1\} \quad \forall\, \mathrm{i}\in \mathrm{I},\mathrm{j}\in \mathrm{J},\mathrm{r}\in \mathrm{R},\mathrm{t}\in \mathrm{T} \tag{13}$$

Equation (14) is included to ensure the CO_2 flow rates in the model being non-negative.

$$F_{ijts} \ge 0 \quad \forall\, \mathrm{i}\in \mathrm{I},\mathrm{j}\in \mathrm{J},\mathrm{t}\in \mathrm{T},\mathrm{s}\in \mathrm{S} \tag{14}$$

4.2. Risk management models

In the two-stage stochastic model, the objective is the expectation of the total cost, which is the average for all the scenarios and thus risk-neutral. While in this context, the variance of the cost for all the scenarios is not addressed. Consequently, the resultant solution would confront high risk with scenario contributing much higher cost than the expectation. To control and manage the risk explicitly, the proposed stochastic model in 4.1 is extended for risk management. In the following sections, three risk metrics, namely, variability index, probability financial risk, and downside risk are implemented to illustrate the effects of the decision-maker's tolerance of risk on the optimal deployment of CCS system under uncertainty.

4.2.1. Variability index

The metric of variability index (also known as 'upper partial mean') is proposed by Ahmed and Sahinidis [44] to take into account the positive difference between the cost of scenario s and the expectation of all scenarios. Compared with the formal variance, the use of variability index avoids the quadratic terms and reduces the complexity. Mathematically, the variability index is defined as shown in equations (15) and (16) [44]:

$$\left\{\begin{array}{l}\Delta_s = \mathrm{TC_s} - \sum_{s\in S} p_s TC_s,\ if\ \mathrm{TC_s} \ge \sum_{s\in S} p_s TC_s \\ \Delta_s = 0, \mathrm{otherwise}\end{array}\right\} \quad \forall\, \mathrm{s}\in \mathrm{S} \tag{15}$$

$$VarIndex = \sum_{s\in S} p_s \Delta_s \tag{16}$$

Equation (15) indicates that if the scenario cost is less than the expectation, Δ_s equals to 0; otherwise, Δ_s is the difference between scenario cost and expectation. Accordingly, the variability index is expressed into the expectation of the positive deviation for all scenarios, as shown in equation (16). It is noticed that the variability index is an asymmetric risk measure that would take account for the scenario having higher cost than the expectation.

The variability index risk management model is formulated as a multiple objective variability index (MO-VI) problem and summarized as follows:

$$MO-VI \begin{cases} \min\ ETC = E[TC_s] = Fixed\mathrm{Cost} + \\ \qquad\qquad \sum_{s\in S} p_s \times Operation\mathrm{Cost}_s \quad (1) \\ \min\ \mathrm{VarIndex} = \sum_{s\in S} p_s \Delta_s \quad (16) \end{cases}$$

s.t. Constraints (2)–(15)

4.2.2. Probabilistic financial risk

The introduction of variance or variability index would generate robust solution in general. However, it might not be sufficient to reduce the risk by variance reduction scheme due to the extreme solution such as the highest cost of all the scenarios. If the decision-maker is risk-averse, he or she might not be willing to take risk for such high cost. To circumvent this problem, an alternative metric named probabilistic financial risk is applied. The definition of this risk metric is proposed by Barbaro and Bagajewicz [45], wherein the probability of the scenario having higher cost than a predefined cost threshold or target Ω is taken into account. A binary variable Δ_s is introduced to indicate whether the

cost is higher than the threshold, which is expressed as equations (17)–(20) [45]:

$$\left.\begin{matrix}\Delta_s = 1, \ \ if\ \mathrm{TC_s}\geq\Omega \\ \Delta_s = 0, \text{otherwise}\end{matrix}\right\} \quad \forall\, s\in S \tag{17}$$

$$\mathrm{ProRisk} = \sum_{s\in S} p_s \Delta_s \tag{18}$$

$$\mathrm{TC_s} \leq \Omega + M \times \Delta_s, \quad \forall\, s\in S \tag{19}$$

$$\mathrm{TC_s} \geq \Omega - M \times (1 - \Delta_s), \quad \forall\, s\in S \tag{20}$$

The binary variable in equation (17) indicates that if the scenario cost is higher than the threshold Ω, Δ_s would be 1; otherwise equals to 0. Therefore, the probability financial risk is the summation of probability times Δ_s for all scenarios. To determine the value of Δ_s, two Big-M constraints in equations (19) and (20) are incorporated for auxiliary, where M refers to an auxiliary constant with large positive value. Equations (19) and (20) indicate that Δ_s must be 1 if TC_s is greater than Ω; otherwise Δ_s must be 0 else constraint (20) would be violated.

The probabilistic financial risk management model appearing as multiple objective probabilistic financial risk (MO-PR) problem is expressed in follows:

$$MO-PR \begin{cases} \min\ ETC = E[TC_s] = Fixed\mathrm{Cost} + \sum_{s\in S} p_s \times Operation\ \mathrm{Cost}_s \quad (1) \\ \min\ \ \mathrm{ProRisk} = \sum_{s\in S} p_s \Delta_s \quad (16) \end{cases}$$

s.t. Constraints (2)–(14); (17); (19)–(20)

4.2.3. Downside risk

To calculate the probabilistic financial risk, a binary variable is required for each scenario. While in the large-scale optimization problem, the introduction of such binary variables would aggravate the difficulty of solving. To avoid trapping into this dilemma, another risk metric named downside risk is applied in this study. Downside risk is also a risk measure focusing on the extremes of the cost distribution [46]. The aim of this metric is to reduce the risk associated with the scenario whose cost is higher than the threshold Ω. Mathematically, the downside risk index is defined as shown in equations (21) and (22) [46]:

$$\left.\begin{matrix}\Delta_s = \mathrm{TC_s} - \Omega, \ \ if\ \mathrm{TC_s}\geq\Omega \\ \Delta_s = 0, \text{otherwise}\end{matrix}\right\} \quad \forall\, s\in S \tag{21}$$

$$\mathrm{DownRisk} = \sum_{s\in S} p_s \Delta_s \tag{22}$$

Equation (21) indicates that if the scenario cost is higher than the threshold Ω, Δ_s would be the positive difference between the scenario cost and threshold Ω; otherwise equals to 0. Thus, downside risk is the summation of the product of probability p_s and Δ_s for all the scenarios.

The downside risk management model is formulated as a multiple objective downside risk (MO-DR) problem and is given below:

$$MO-DR \begin{cases} \min\ ETC = E[TC_s] = Fixed\mathrm{Cost} + \sum_{s\in S} p_s \times Operation\mathrm{Cost}_s \quad (1) \\ \min\ \ \mathrm{DownRisk} = \sum_{s\in S} p_s \Delta_s \quad (22) \end{cases}$$

s.t. Constraints (2)–(14); (21)

4.2.4. Multi-objective problem solution procedure

As indicated, the three risk management models all have two objectives: minimize the ETC meanwhile minimize the risk metric. To generate the set of optimal solutions representing the trade-offs between economic and risk criteria, ε-constraint approach [42] is implemented. The multi-objective optimization problem can be transformed as single-objective problem, which is mathematically formulated as model (P1).

(Model P1)

$$\min\ ETC(x, y, z)$$

Subject to:

$$Risk\mathrm{Metric}(x, y, z) \leq \varepsilon$$

$$\mathrm{h}(x, y, z) = 0$$

$$\mathrm{g}(x, y, z) \leq 0$$

where ε is the auxiliary parameter. In this paper, the economic objective is optimized while incorporating the risk objective as additional constraint that is forced to be lower than the epsilon parameter. The Pareto solutions are obtained by setting different values of the parameter ε, which includes three steps. First, the minimum ETC is obtained by regarding the risk metric as parameter. This entails performing the single objective optimization which in turn yields the largest risk metric ($\varepsilon_{\max}$). The second step is to minimize the risk metric ($\varepsilon_{\min}$), and in turn yields the largest ETC. The last step is to retain ETC as the main objective while setting the parameter ε to the discrete values between upper ($\varepsilon_{\max}$) and lower ($\varepsilon_{\min}$) bounds. Finally, the approximation to the Pareto curve showing relationship between the economic and risk objectives can be obtained.

5. Case study: CCS in northeast China

In this section, a realistic case study from Northeast China is investigated to demonstrate the capabilities of the two-stage stochastic programming model (Case I: risk-neutral) and the extended risk management models (Case II: risk-averse). Considering the uncertainty of carbon tax, the purpose is to make the optimal CCS network plan for energy insensitive industries with risk preferences of decision-makers. The results for these two cases are presented and discussed. All the cases are solved to optimality using GAMS/CPLEX 12.5 [47] on a Desktop with an Intel (R) Core (TM) i5 processor at 3.20 GHz, and 16 GB of RAM. The optimality tolerance for all the cases in this work is set to be 0.

5.1. Basic data

This case study puts focus on the Northeast region of China, including Heilongjiang province, Jilin province, Liaoning province and five league cities in eastern Inner Mongolia. There are reasons to select this region as study object. First, Northeast China is a significant heavy industries base for China, locating many facilities of power sectors and manufacturing sectors [48]. These sectors accounts for the majority of CO_2 emission in China. Second, there are many oil fields and deep saline reservoirs could be used for carbon storage in this region [49]. Third, the government has launched several rounds of initiatives to reduce CO_2 emission as part of the Northeast revitalization strategy. These favorable conditions make it possible for large-scale application and commercialization of CCS system.

The major data is adapted from the authors' previous work [50], which consists of six power plants as emission sources and three geological reservoirs as sinks. The main characteristics of sources and sinks are shown in Table 1 and Table 2. For all capture facilities, capture efficiency factor is assumed to be 0.9 [29], meaning that up to 90% of the CO_2 emission would be captured and stored. This assumption is reasonable for most realistic scenario and would drastically reduce complexity of the problem [26]. The capital cost for the capture facility is deduced depending on the feed gas flowrate, composition and capture technology, as equation (23) [26] shows:

$$CIC = \alpha + (\beta x^n + \gamma) \times F^m \tag{23}$$

where F is the total feed flow rate (mol/s), x is the feed CO_2

Table 1
Power plants: CO2 emission, operation time, capital and O&M cost for capture facility. Sources [22,50].

Plant	Operation time (y)	Composition	Emission (Mt/y)	Capital cost ($)	O&M cost ($/t)
1	0–30	14%	9.20	250.17×10^7	30.28
2	0–40	18%	6.15	173.92×10^7	23.78
3	5–30	11%	5.90	182.82×10^7	23.19
4	10–40	17%	5.86	168.80×10^7	23.10
5	0–40	15%	5.58	165.78×10^7	22.43
6	10–50	15%	5.00	151.81×10^7	21.12

composition and $\alpha\ \beta\ \gamma\ n\ m$ are all estimated parameters.

Among the existing CO_2 transportation alternatives, pipeline is considered as the most mature approach for large volume CO_2 transportation. Various available pipeline cost models have been reviewed and compared by Knoope et al. [51]. In this study, the weight-based model is employed for cost evaluation of pipeline, as expressed in equations (24)–(26) [51].

$$W_{material} = \pi \cdot \rho_{material} \cdot L \cdot (\mathrm{r} - t_{\mathrm{p}}) \tag{24}$$

$$t_{\mathrm{p}} = \frac{P_{\max} \times \mathrm{r}}{2 \times S_{\mathrm{p}} \times F_f \times E_{\mathrm{f}}} \tag{25}$$

$$I = W_{material} \times \frac{C_p}{f_m} \tag{26}$$

wherein $W_{material}$ in equation (24) is the weight of the pipeline with certain material and is calculated according to its density ($\rho_{material}$), length (L), diameter (r) and thickness (t_p). The wall thickness of the pipeline is obtainable through equation (25) with knowing the maximum operation pressure ($P_{\max}$), diameter, minimum yield stress (S_p), design factor (F_f) and longitudinal joint factor (E_f). Such that the total capital cost can be achieved with equation (26), by dividing the weight times material price (C_p) by a fraction factor (f_m). The pipeline diameters to be considered in the model should not be arbitrary but in discrete values of standard size. Therefore, this study uses the data presented by Yue et al. [52] to obtain the possible diameter values for various flow rate ranges; and then employ the weight-based model to obtain the capital cost with certain diameter. The related pipeline information is summarized in Table 3 and the distance data between sources and sinks are given in Table 4.

5.2. Case implementation

Case I: Stochastic Model is implemented to make the best CCS planning towards the minimum ETC if the decision-maker is risk-neutral. The operating horizon is subdivided into 50 time periods with 1 year lasting for each, and the uncertainty of carbon tax is concurrently investigated in terms of different scenarios. This work assumes it follows the normal distribution with mean value of $ 50/t CO_2, deviation of $ 10/t CO_2 though the historical record varies a lot. This model could serve as a decision support tool for the government policy maker to determine the appropriate ranges of carbon tax, or for the industrial emitters to decide the most economical strategy for CO_2 reduction.

Case II. Risk management models with three different risk metrics are implemented if the decision-maker is risk-adverse. In this case, all the relevant data is same to case I. The target of this case is to minimize ETC and risk metric simultaneously within the CCS deployment planning. The formulations of the three risk management metrics are employed respectively to reduce the risk imposed by the uncertainty. Thus, a multi-objective optimization problem is formulated and the optimal solutions with each risk management metric will yield Pareto curves. A wide variety of Pareto-optimal solutions can be obtained for selection according to decision-maker's preference.

The initial step in both cases is to generate adequate scenarios representing the varied carbon tax price. Before performing the models, stochastic convergence analysis is needed to determine the number of scenarios. Fig. 3 shows the convergence of the expected value of total cost with the number of scenarios. It can be found that ETC is steady enough to accurately approximate the expectation when scenarios are no less than 450. Thus 450 scenarios of different carbon taxes are generated with Monte Carlo for CCS planning.

Table 3
Pipelines: diameter, capacity bound, capital and O&M cost. Sources [52].

Diameter (inches)	Flowrate lower bound (Mt/y)	Flowrate upper bound (Mt/y)	Capital Cost ($/km)	Yearly O&M Cost ($/km)
8	0.43	1.13	42.44×10^3	1697.87
12	0.90	3.25	95.50×10^3	3820.21
16	2.60	6.86	169.79×10^3	6791.48
20	5.49	12.26	265.29×10^3	10611.69
24	9.81	19.69	382.02×10^3	15280.84

Table 4
Distances between sources and sinks (km). Sources [50].

Sink			
Source	MA	MB	MC
1	285.0	483.9	447.4
2	273.6	648.0	653.9
3	626.4	363.3	218.3
4	166.3	616.4	820.9
5	921.8	701.9	245.3
6	173.1	412.9	787.4

6. Results and discussions

In this section, the obtain results for the two cases are presented in Section 6.1 and 6.2 respectively.

6.1. Results for two-stage stochastic programming model

Carbon tax price will influence decision-maker's attitude for participating in the CCS system, so this section will evaluate the interaction between carbon tax policy and construction of CCS system with considering the uncertainty of the carbon tax price. The two-stage stochastic programming model with 450 scenarios is solved. Noted that the carbon tax for each scenario is unique. Through model optimization, the most desired planning result is obtained at the expected total cost of $ 643.46×10^8, which is the minimum summation of the costs for CCS capital cost ($ 41.57×10^8), CCS O&M cost ($ 108.11×10^8)

Table 2
Potential reservoirs: type, injection limit, start time, capital and O&M cost for storage facility. Sources [22,50].

Sink	Type	Injection limit (Mt/y)	Start time (y)	Capital cost ($)	O&M cost ($/t)
MA	Oil Field	28	0	40×10^7	12
MB	Saline formation	20	10	66×10^7	8
MC	Saline formation	30	15	50×10^7	5

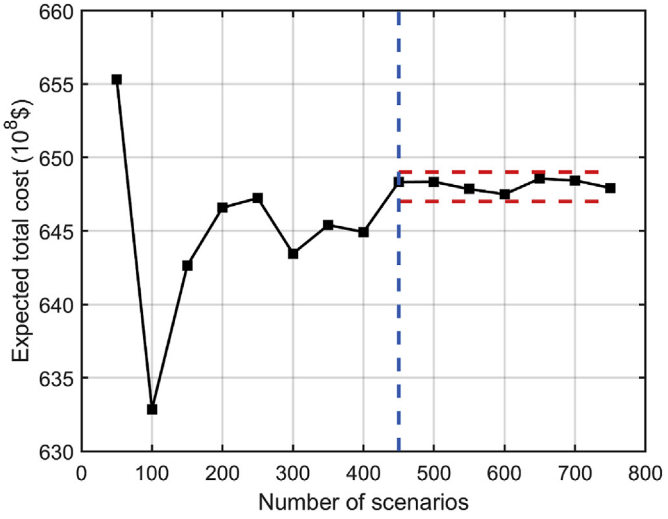


Fig. 3. Convergence of ETC with the number of scenarios for case study.

and carbon tax surcharge ($ 493.78 × 10^8). While locating to each scenario, ETC spreads from $ 344.94 × 10^8 to $ 954.91 × 10^8. It can be concluded that when the price on carbon tax is increased, the emission sources prefer to implement a CCS network and also the network becomes more complex, thereby reducing the surcharge paid for the CO_2 not captured. The network structure is presented in Table 5. As shown, the CO_2 emission from plant 3 and 6 are captured and subsequently stored into storage site MB. In network detail, CO_2 from plant 3 and 6 are transported to storage site MB via 16-inch pipelines. This network is shared by all concerned scenarios however the relevant stream flowrates are different for each as result of the varying of carbon tax.

The cost distribution is shown in Fig. 4. According to the relationship between carbon tax and O&M cost, the following two critical points for carbon tax can be found. If the carbon tax price is lower than $ 27/t, most scenarios tend to pay the surcharge. While the carbon tax price is higher than $ 50/t, more scenarios prefer CCS and the number of scenarios are gradually increased. As for the cost distribution, it spreads widely and is likely to cause risk since having scenario cost much higher than the average. If the decision-maker is risk-averse, the solutions obtained from two-stage stochastic programming model (i.e., risk-neutral solution) will not be satisfied or even infeasible. In order to control and manage the associated risk imposed by uncertainty, risk management is needed.

6.2. Results for risk management models

6.2.1. Results for variability index management

Variability index as mentioned before is a risk metric penalizing the scenarios' cost higher than the expectation. This section evaluates of the trade-off between ETC and variability index. A multi-objective optimization scheme considering these two objectives is conducted by solving the MO-VI model through ε-constraint approach. First, the two extreme solutions, namely, minimal ETC solution and the minimal variability index solution are determined at $ 642.64 × 10^8 and $ 722.09 × 10^8 respectively. On these bases, 5 evenly distributed ε-parameter values are defined between the lower and upper bounds of the variability index objective, which is from $ 0 to $ 38.13 × 10^8 in this case. The resulting ETC versus variability index set of Pareto-optimal solutions are presented in Fig. 5, with the abscissa representing variability index, the ordinate indicating ETC. The obtained 7 Pareto-optimal solutions in total constitute the approximation to the Pareto curve. The solutions above this curve are suboptimal, below are infeasible, locating on are considered optimal. The points on the right of the curve tend to minimize the total expected cost, while the ones on the left tend to pursue a more risk-averse solution. Here, it can be found that with the variability index increasing from $ 0 to $ 38.13 × 10^8, ETC decreases from $ 722.09 × 10^8 to $ 642.64 × 10^8, indicating the contradictory between the two objectives. More than this, the cost breakdown for the Pareto-optimal solutions can be presented as well. As shown in Fig. 6, the major cost sources from O&M cost of CCS system. Overall, as the variability index decreases, the contribution of O&M cost increases and the associated surcharge cost decreases.

The comparison of cost distribution before and after managing the variability index is shown in Fig. 7 (ε (4) risk level). It can be concluded that compared with risk-neutral solution, the spread of the cost for all the scenarios is narrowed after managing the risk, but the expectation shifts to higher cost. On the other hand, to reduce the variability index, the probability of having cost around $ 700.00 × 10^8 increases significantly from 11.33% to 56.00% since the expectation of this Pareto solution is $ 663.50 × 10^8. Also, the risk-neutral solution has 5.20% probability to have scenario cost higher than $ 800.00 × 10^8, while after variability index risk management, the new solution has only 2.00% probability to exceed that cost (greater than $ 750.00 × 10^8), which means that the risk of higher cost has been reduced after variability index management at the expensive of reducing the probability of having scenario whose cost is much lower (less than $ 550 × 10^8).

6.2.2. Results for probabilistic financial risk management

Probabilistic financial risk focuses on the positive difference between the cost for each scenario and a predefined threshold Ω established by the decision-maker. The threshold represents the maximum cost value the decision-maker can tolerate. In this case, Ω is set to $ 710 × 10^8. Similar with the variability index management model, a multi-objective optimization considering the minimal probabilistic financial risk and minimal ETC is conducted by solving the MO-PR model via the ε-constraint approach. The Pareto-optimal curve for this case is presented in Fig. 8. As shown, there are seven Pareto-optimal solutions. The probabilistic financial risk increases and ETC decreases as expected. When probabilistic financial risk is minimized, the obtained CCS configuration implies the expected total cost of $ 651.14 × 10^8 and the probabilistic financial value of 0.12. Conversely, when ETC is minimized, the cost reaches $ 639.11 × 10^8 and the risk increases to 0.20. The cost breakdown for the Pareto-optimal solutions can be obtained as well. As shown in Fig. 9, the primary cost comes from the O&M cost of CCS system and surcharge cost. With the decrease of risk, it can be concluded that the carbon tax cost tends to decrease, and O&M cost contributes equally with tax cost finally.

Fig. 10 is the comparison of cost distribution before and after managing the probabilistic financial risk (ε (3) risk level). As shown, it can be concluded that the spread of ETC is slightly narrowed after managing the probabilistic financial risk and all the cost values obtained from each scenario are approximated to not exceeding the threshold established by the decision-maker ($ 710.00 × 10^8). From the perspective of achieving target, after probabilistic financial risk management, it has only 7.00% probability to have the higher cost than $ 710.00 × 10^8. Compared with the results obtained by solving MO-VI model, it can be concluded that the results from solving MO-PR model still have the probability of getting scenarios whose cost is lower than $ 550.00 × 10^8 while in MO-VI model that probability is almost 0.

Table 5
Capture facilities and pipelines installed.

Capture facility	Connection	Diameter (inches)
Power plant 3	MB	16
Power plant 6	MB	16

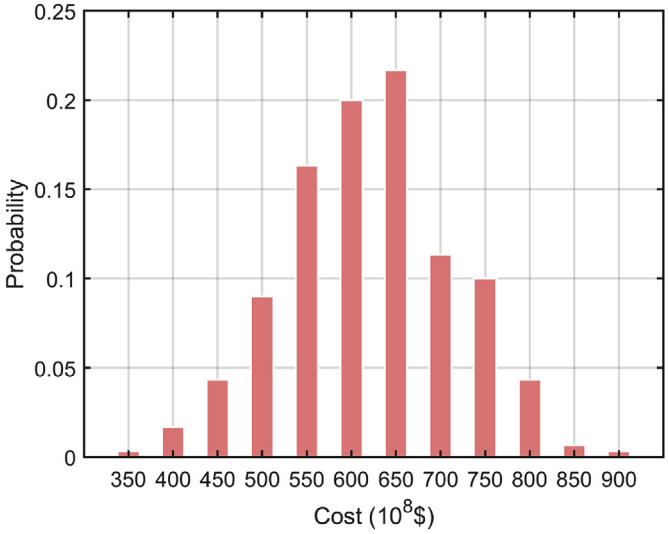


Fig. 4. The histogram of results for stochastic programming model (risk-neutral solution).

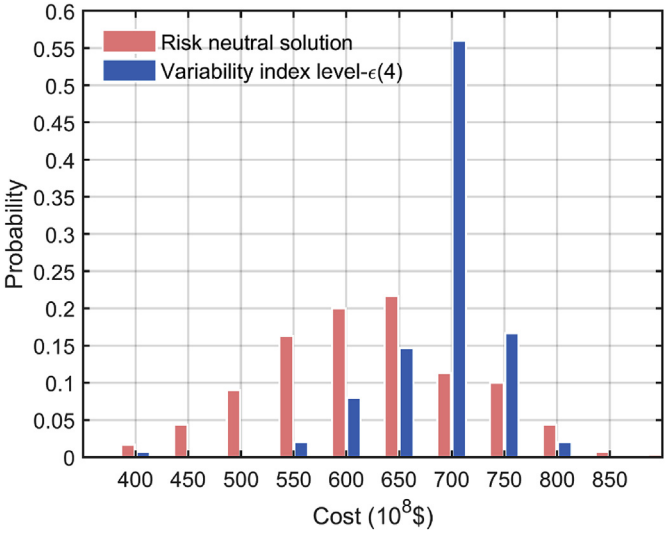


Fig. 7. The histograms of cost distribution before and after variability index management-ε (4).

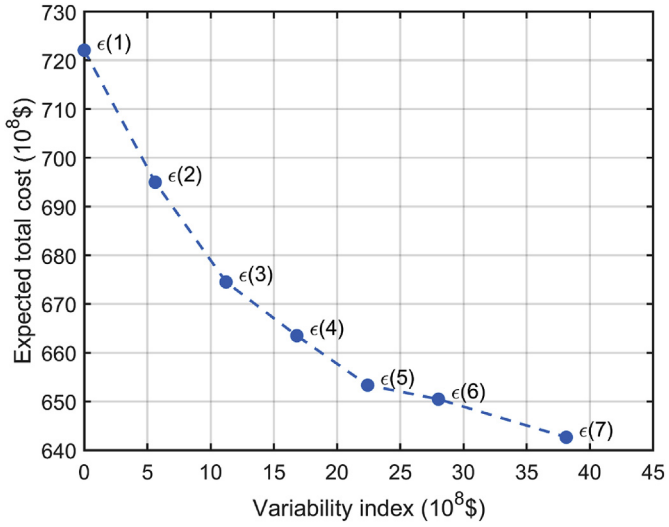


Fig. 5. The Pareto-optimal curve for variability index and ETC.

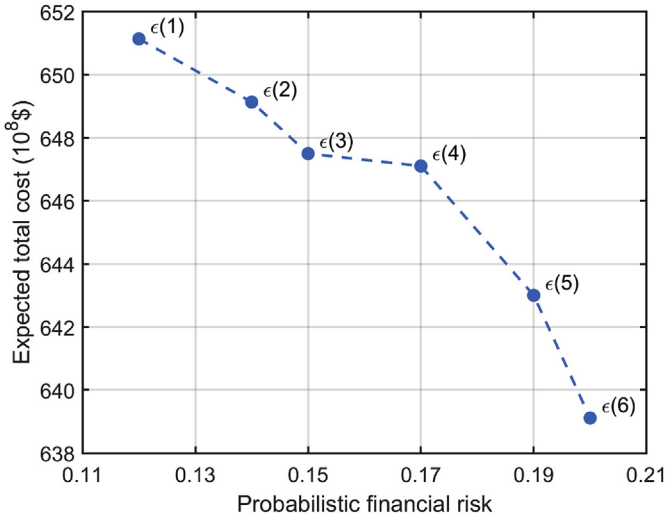


Fig. 8. The Pareto-optimal curve for probabilistic financial risk and ETC.

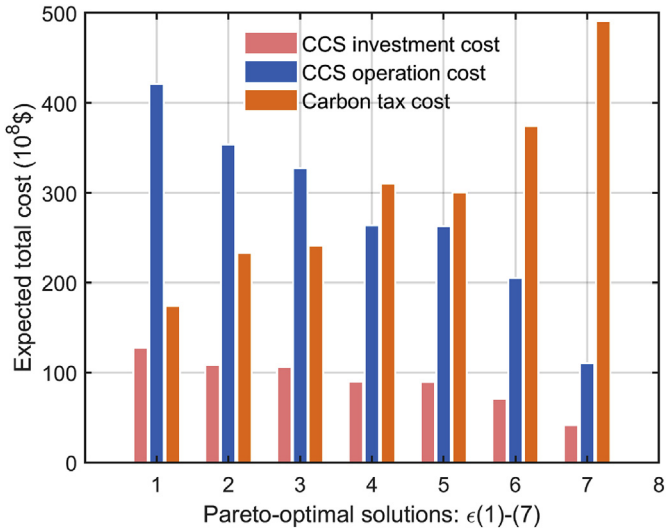


Fig. 6. The cost breakdown for the Pareto-optimal solutions of variability index and ETC.

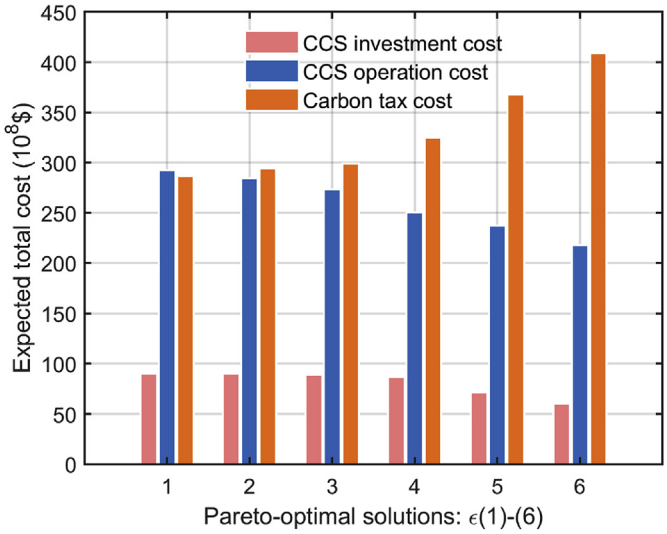


Fig. 9. The cost breakdown for the Pareto-optimal solutions of probabilistic financial risk and ETC.

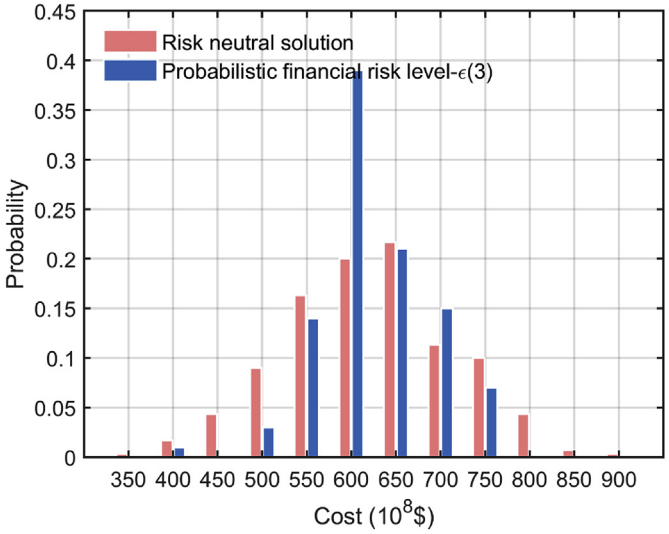


Fig. 10. The histograms of cost distribution before and after probabilistic financial risk management-ε (3).

6.2.3. Results for downside risk management

The downside risk metric is similar to probabilistic financial risk metric, but it reduces the number of binary variables. In the calculation of the downside risk, the threshold value Ω is also set to \$ 710.00×10^8. By solving the MO-DR model via the ε-constraint approach, an approximated Pareto curve as depicted in Fig. 11 is obtained, where the abscissa denotes the downside risk, and the ordinate denotes ETC of the CCS system. The lower and upper bounds of downside risk can be obtained through the two extreme solutions first: the minimal downside risk solution and the minimal ETC solution; then, according to the downside risk ranges, the value of ε-parameter is evenly distributed to 4 values between the lower and upper bounds of the downside risk objective, i.e., from \$ 15.69×10^8 to \$ 24.81×10^8. Once again as expected, it can be found that as the downside risk increases from \$ 13.38×10^8 to \$ 27.09×10^8, the ETC decreases from \$ 663.07×10^8 to \$ 642.64×10^8. As shown in Fig. 12, the primary cost comes from the O&M cost of CCS system, which contributes most among all the Pareto optimal solutions.

The histograms of cost distribution before and after managing the downside risk (ε (3) risk level) are shown in Fig. 13. Also, after managing the downside risk, the probability of having high cost greater than \$ 710.00×10^8 is only 10.00%. It can be found that the shapes of the Pareto curve in this case are almost the same with that in probabilistic financial risk model. This is because the main ideas behind these two metrics are identical, which are all focus on the parts that greater than the predefined threshold. However, as for the cost distribution, it can be found that downside risk metric prefers to CCS operation while probabilistic risk metric prefers to surcharge.

6.3. Comparisons of the risk metrics

By analyzing the results obtained from the stochastic programming model and the three risk management models, it can be found that risk management always entails an increase of ETC in the CCS planning for risk reduction. Nevertheless, inclusion of risk metrics to the optimal deployment of CCS system under uncertainty can yield more robust designs and each one has advantages and disadvantages.

For the metric of variability index, it can be concluded that the variability index is appropriate if the decision-maker prefers a solution near the expectation. This risk metric is in linear representation, so the programming problem can be solved more easily. Compared with the formal variance, variability index seems more reasonable since the scenarios with lower costs than expectation are not penalized, but it should also be noted that it is difficult for the metric to cover the low-cost scenarios because the spread of the costs for all the scenarios is relatively narrowed.

Regarding probabilistic financial risk and downside risk metrics, these two are related since they both involve establishing a threshold predefined by the decision-maker. The risk is measured by the expectation of scenarios whose costs are higher than the threshold. However, the threshold value is subjective and often difficult to explicitly predefined. While between the two metrics, downside risk is believed easier to be solved than probabilistic financial risk as fewer binary variables are included.

The inclusion of the most appropriate risk metric relies on the willing of decision-maker. If the decision-maker is more concerned with the variability of the results, the variability index should be used. This risk measure only penalizes the solutions higher than the expectation. While if he/she cannot tolerate the cost higher than a specific threshold, probabilistic financial risk or downside risk are suitable, because they concentrate more on the threshold.

7. Practical implications

The findings of this study highlight several important implications about the importance of carbon tax uncertainty in the large-scale deployment of CCS for Northeast China. First, this study identifies the relationship between the two reduction options: carbon tax and carbon capture and storage. This could serve as a decision support tool for the government policy-maker to determine the appropriate ranges of carbon tax, or for the industrial emitters to decide the most economical strategy for CO_2 reduction. Second, although the proposed risk management framework will push up the expected total cost of CCS planning, inclusion of risk metrics can yield more robust designs. In this way, a wide variety of Pareto-optimal solutions can be obtained for selection according to decision-maker's preference. The case of Northeast China study reveals that the huge cost for the cleaner production with CCS has hindered its deployment. A tax concession policy (higher than \$ 50/t) or low-interest loans can somehow achieve the promotion. Finally, this study can be used as a base case before the implementation of nationwide CCS system.

8. Conclusions

The motivation for this study is to provide quantitative information for policy-makers to make an efficient decision on the issue of CCS planning under uncertainty arisen from fluctuation in price of carbon

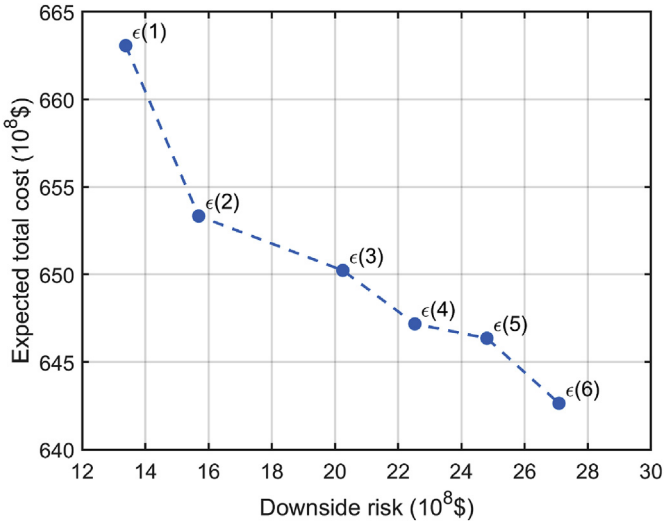


Fig. 11. The Pareto-optimal curve for downside risk and ETC.

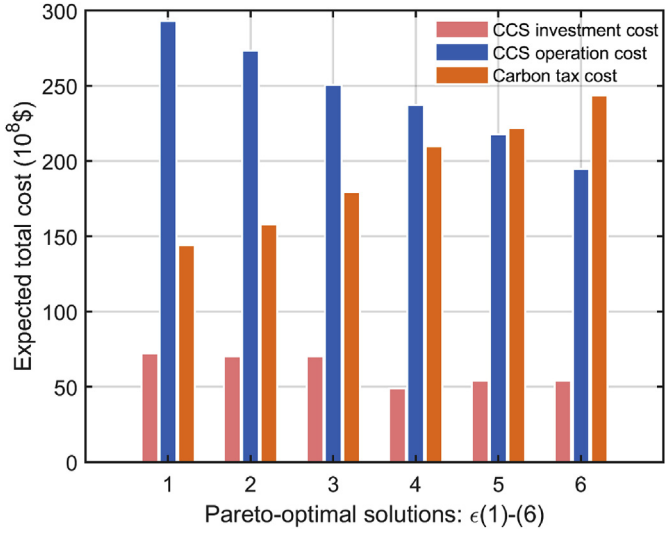


Fig. 12. The cost breakdown for the Pareto-optimal solutions of downside risk and ETC.

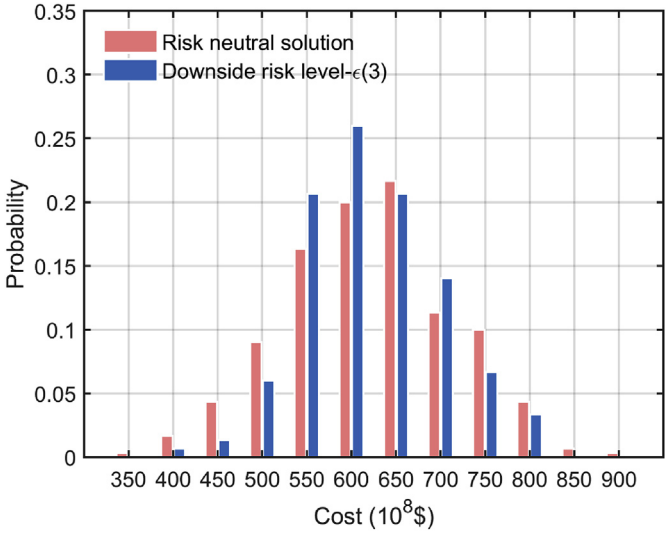


Fig. 13. The histograms of cost distribution before and after downside risk management-ε (3).

tax. For this purpose, this study has developed a two-stage stochastic MILP model that aims to minimize the expectation of costs for carbon capture, pipeline cost for transporting, storage cost for injecting and the surcharge for carbon tax. Based on such economic incentive condition, industrial emitters can choose to involve in CCS or not, however, yielding a risk-neutral solution. To assess the risk imposed by the uncertainty, three risk metrics (variability index, probability financial risk, and downside risk) are extended as objective functions and applied separately with expected total cost in multi-objective optimization. To show the trade-offs between ETC and risk, approximation of the Pareto-optimal curves is developed through the application of the ε-constraint method for each risk metric.

A case study from Northeast China is presented to show the application of the proposed methodology in this research study. Numerical results have been reported that the goal is to be able to determine the best CCS network, under uncertain carbon tax price condition. It has been found that such decision completely depends on the desired tolerance of risk level. The risk-neutral solution provides policy recommendations that if the carbon tax price is lower than $ 27/t, the emitter tends to pay the surcharge and therefore will not promote the application of CCS. However, if the price is higher than $ 50/t, most scenarios prefer CCS. This practical implication could help the government to identify the appropriate ranges of carbon tax price for the promotion of CCS. In risk-averse cases, the use of three risk metrics proves to be enough to assess the risk at the expense of increasing ETC where variability index focuses on variance reduction (from $ 0 to $ 40×10^8), probability financial risk (from 0.12 to 0.20) and downside risk (from $ 13.38×10^8 to $ 27.09×10^8) metrics focus on the predefined target. It can be concluded that both the risk-averse cases prefer CCS with the tolerance of risk decreased. Finally, it can be stated that the proposed risk management framework is useful to provide quantitative risk information during regionwide planning stage. Otherwise, the risk-neutral solution may confront high levels of risk and may be infeasible in practice.

On basis of this study, future work will be launched to evaluate the other key factors such as the reservoir characteristics since large uncertainties exist for the injection and storage of CO_2. Although this work has investigated the influence of carbon tax uncertainty, the final performance of the storage reservoir is unclear until injection has started. These two kinds of uncertainties would generate great impacts on the success of large-scale CCS deployment. Moreover, environmental issue is also of great significance for the optimal planning of a sustainable CCS network.

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